

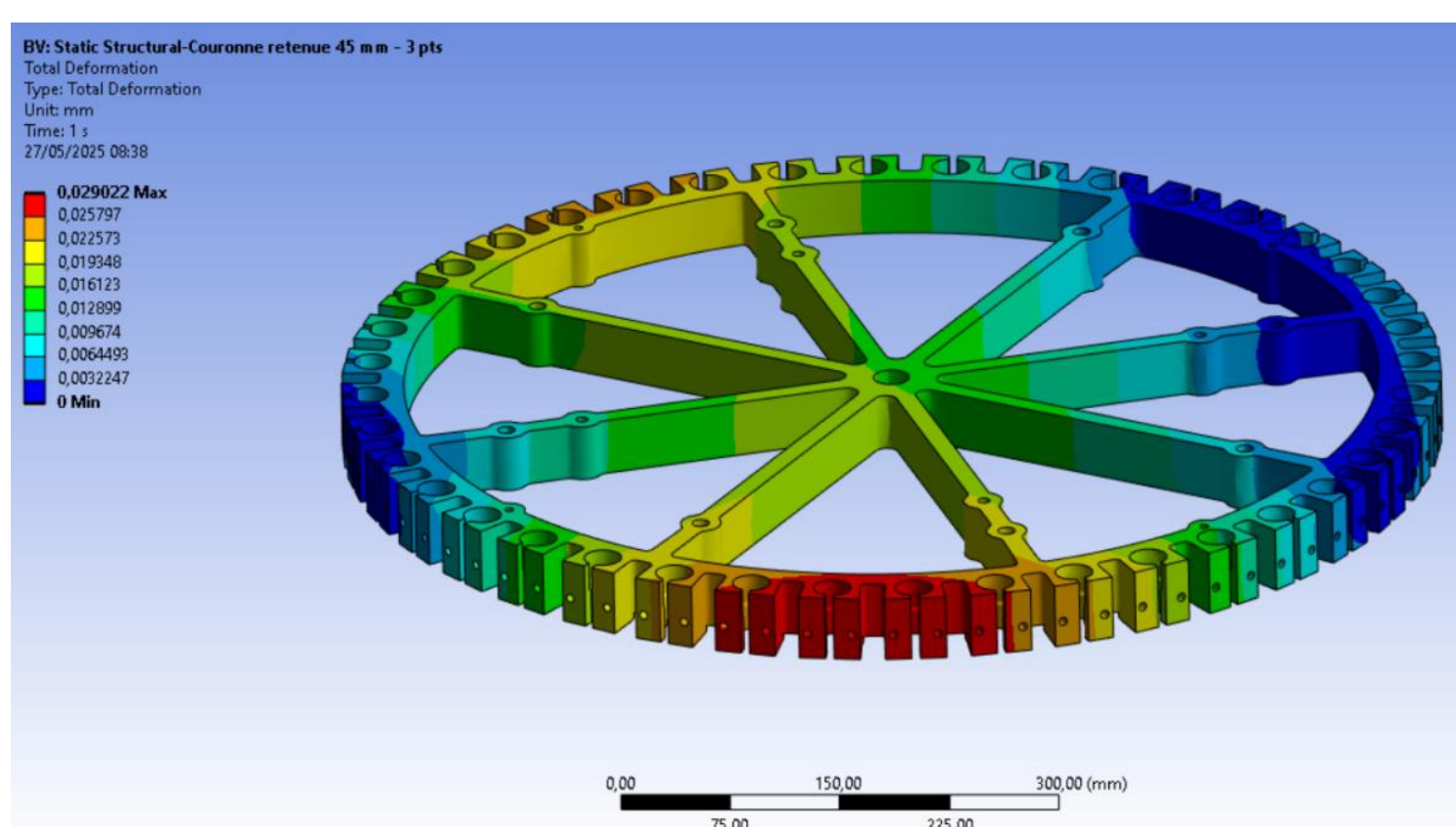
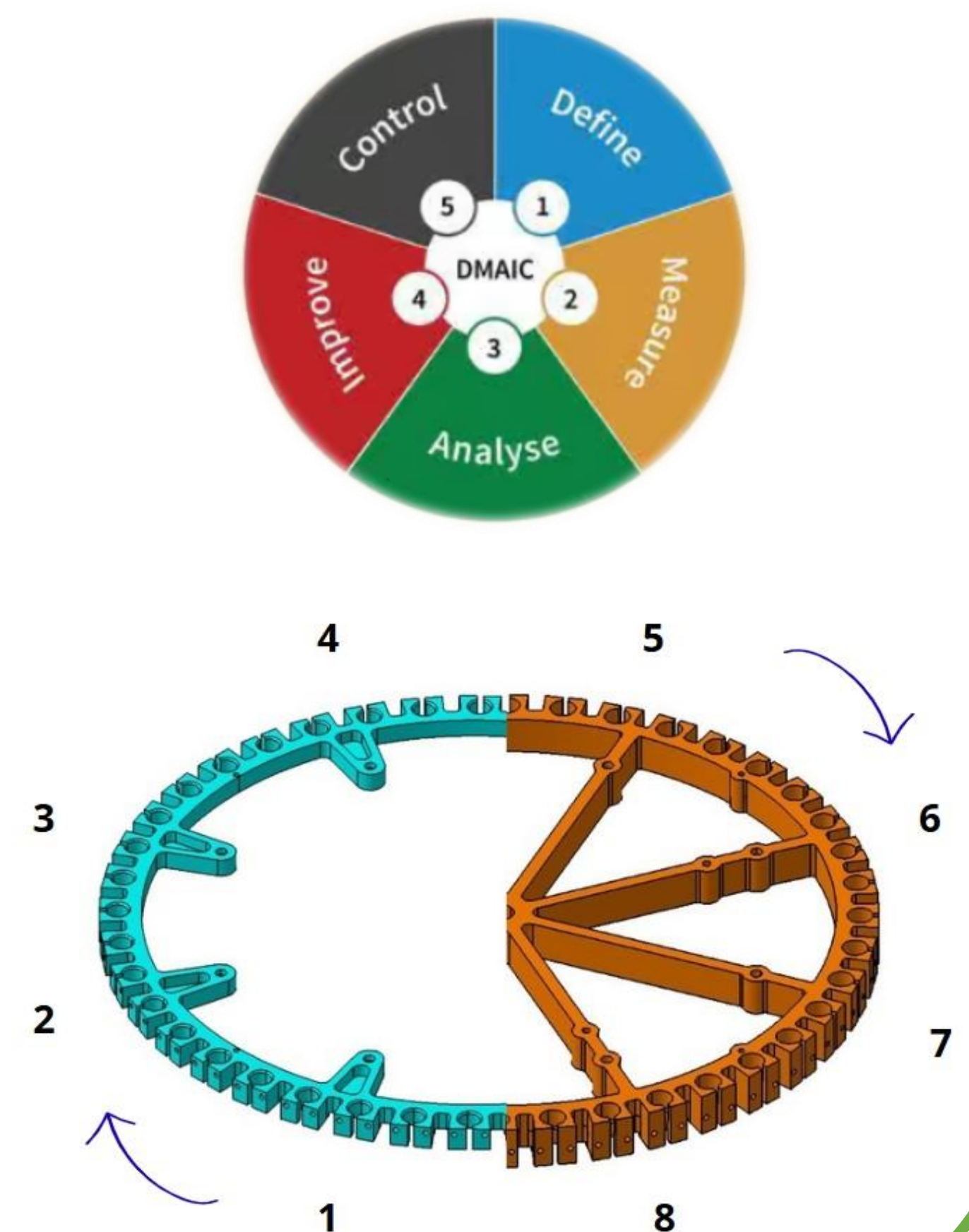
Improvement of the Three-Dimensional Inspection Tooling

This study focuses on improving a dimensional inspection fixture for a sealing flange support on the LEAP-B1 engine, where the current tool **fails to meet the 0.025 mm flatness** requirement.

Crown Tooling: Symbolising the DMAIC Iterative Process

Eight Design Steps Mirroring the Fixture's Eight Spring Supports

1. Measurement of the current fixture (CMM + FEA)
2. Identification of deformation root causes
3. Definition of dimensional & ergonomic constraints
4. CAD modelling with parametric variations (CATIA)
5. Geometric enhancements (ribs, rings, extra thickness)
6. Finite Element Analysis in ANSYS Workbench
7. Performance-to-mass trade-off evaluation
8. Final selection and validation with stakeholders



Criteria	Old Crown	New Crown
Deformation	0.25 mm	0.04 mm
Mass	11.37 kg	30.2 kg

- ✓ - 80% of deformation gain
- ✓ Limited mass
- ✓ Compatible with manual handling

The final design combines reduced deformation, limited mass, and manufacturability, offering a **validated solution for dimensional control** of flexible aeronautical parts. **Further work** includes topology optimisation, fatigue testing, and standardisation.

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